Physics

1. A rectangular coil (Dimension 5 cm × 2.5 cm) with 100 turns, carrying a current of 3 A in the clock-wise direction, is kept centered at the origin and in the X-Z plane. A magnetic field of 1 T is applied along X-axis. If the coil is tilted through 45° about Z-axis, then the torque on the coil is:

A. 0.55 Nm B. 0.38 Nm

C. 0.27 Nm D. 0.42 Nm

Ans. D.

Sol: The torque on the coil is given by $\tau = M \times B$

$= NIA\ B \sin\theta$

$= 100 \times 3 \times 5 \times 2.5 \times 10^{-4} \times 1 \times \sin 45^o$

$= 0.27$ Nm

2. If 'M' is the mass of water that rises in a capillary tube of radius 'r', then mass of water which will rise in a capillary tube of radius '2r' is:

A. 4 M B. 2 M

C. M D. $\frac{M}{2}$

Ans. B.

Sol:

Force inside the capillary tube is

$2\pi T = Mg \Rightarrow M \propto r$

when r' = 2r, then

M' = 2M

3. Taking the wavelength of first Balmer line hydrogen spectrum (n = 3 to n = 2) as 660 nm, the wavelength of the 2nd Balmer line (n = 4 to n = 2) will be:

A. 388.9 nm B. 488.9 nm

C. 642.7 nm D. 889.2 nm

Ans. B.

Sol:

The wavelength of first hydrogen spectrum is given

$$\frac{1}{\lambda} = R\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

$$\frac{1}{660} = R\left(\frac{1}{2^2} - \frac{1}{3^2}\right) = \frac{5R}{36}$$

$$\frac{1}{\lambda} = R\left(\frac{1}{2^2} - \frac{1}{4^2}\right) = \frac{3R}{16}$$

$$\frac{\lambda}{600} = \frac{5/36}{3/11}$$

$$\lambda = 488.88$$

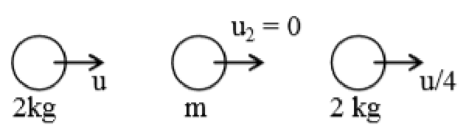
4. A body of mass 2 kg makes an elastic collision with a second body at rest and continues to move in the original direction but with one fourth of its original speed. What is the mass of the second body?

A. 1.8 kg B. 1.2 kg

C. 1.0 kg D. 1.5 kg

Ans. B.

Sol:



From linear momentum conservation**,**

$$v_1 = \frac{2m_2u_2 + u_1(m_1 - m_2)}{m_1 + m_2}$$

$(\because v_1 = \frac{u}{4})$

$$\frac{u}{4} = \frac{0 + u(2 - m)}{2 + m}$$

$$\frac{u}{4} = \frac{u(2 - m)}{2 + m}$$

$$\frac{\cancel{u}}{4} = \frac{\cancel{u}(2 - m)}{2 + m}$$

$$2 + m = 8 - 4m$$

$$5m = 6$$

$$m = \frac{6}{5}$$

$$m = 1.2 \text{ kg}$$

5. A stationary horizontal disc is free to rotate about its axis. When a torque is applied on it, its kinetic energy as a function of θ, where θ is the angle by which it has rotated, is given as $k\theta^2$. If its moment of inertia is I then the angular acceleration of the disc is:

A. $\frac{2k}{I}\theta$ B. $\frac{k}{2I}\theta$

C. $\frac{k}{I}\theta$ D. $\frac{k}{4I}\theta$

Ans. A.

Sol:

Given, kinetic energy = $k\theta^2$

We know that, kinetic energy of a rotation body about its axis = $\frac{1}{2}I\omega^2$

Where, I is moment of inertia and ω is angular velocity.

$\Rightarrow \quad \frac{1}{2}I\omega^2 = k\theta^2$

$\frac{1}{2}I\omega^2 = k\theta^2$

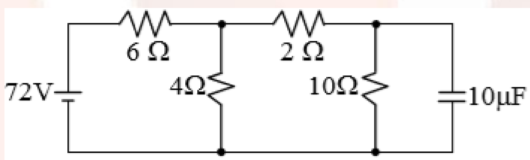
By differentiate

$\frac{1}{2}I \times 2\omega \frac{d\omega}{dt} = 2k\theta \frac{d\theta}{dt}$

$I\omega \frac{d\omega}{d\theta} = 2k\theta$

$\alpha = \frac{2k\theta}{I}$

6. Determine the charge on the capacitor in the following circuit:



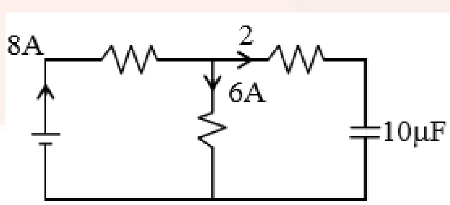
A. 200 μC B. 2 μC

C. 10 μC D. 60 μC

Ans. A.

Sol: Total Resistance = 9

Total current = 8 A

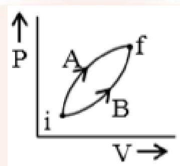


Voltage 10Ω = 20V

q = CV

= 10 μF × 20 = 200 μC

7. Following figure shows two processes A and B for a gas. If ΔQ_A and ΔQ_B are the amount of heat absorbed by the system in two cases, and ΔU_A and ΔU_B are changes in internal energies, respectively, then:



A. $\Delta Q_A > \Delta Q_B$, $\Delta U_A > \Delta U_B$

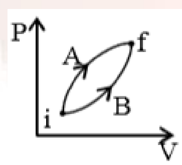
B. $\Delta Q_A < \Delta Q_B$, $\Delta U_A < \Delta U_B$

C. $\Delta Q_A = \Delta Q_B$; $\Delta U_A = \Delta U_B$

D. $\Delta Q_A > \Delta Q_B$, $\Delta U_A = \Delta U_B$

Ans. D

Sol:



Since, the initial and final state are same in both process, so,

So, $dU_A = dU_B$

$dW_A > dW_B$ (Area)

So, $dQ_A > dQ_B$ $(dQ = dU + dW)$

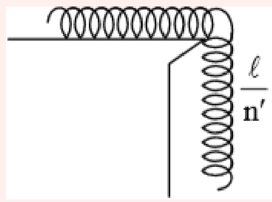
8. A uniform cable of mass 'M' and length 'L' is placed on a horizontal surface such that its $\left(\frac{1}{n}\right)^{th}$ part is hanging below the edge of the surface. To lift the hanging part of the cable upto the surface, the work done should be:

A. $\frac{2MgL}{n^2}$ B. nMgL

C. $\frac{MgL}{2n^2}$ D. $\frac{MgL}{n^2}$

Ans. C.

Sol:



Given, mass of the cable is M.

So, mass of $\frac{1}{n}$th part of the cable, i.e.,

hanged part of the cable is = M/n …(i)

Now, center of mass of the hanged part will be its middle point.

So, its distance form the top of the table will be L/2n.

Work done against gravity = mgh

$= \frac{m}{n} g \frac{\ell}{2n}$

$= \frac{mg\ell}{2n^2}$

9. An HCl molecule has rotational, translational and vibrational motions. If the rms velocity of HCl molecules in its gaseous phase is $\bar{v}$, m is its mass and k_B is Boltzmann constant, then its temperature will be:

A. $\frac{m\bar{v}^2}{3k_B}$ B. $\frac{m\bar{v}^2}{6k_B}$

C. $\frac{m\bar{v}^2}{7k_B}$ D. $\frac{m\bar{v}^2}{5k_B}$

Ans. (Bonus)

Sol:

Energy of molecules $= \frac{f}{2} k_B T$

For rotational, translational and vibrational motions $\frac{1}{2} m\bar{v}^{2} = \frac{6}{2} k_B T$

$T = \frac{m\bar{v}^{2}}{6k_B}$

10. A single Acosωt is transmitted using $\upsilon_0 \sin\omega_0 t$ as carrier wave. The correct amplitude modulated (AM) signal is:

A. $(\upsilon_0 + A)\cos\omega t \ \sin\omega_0 t$

B. $\upsilon_0 \sin\omega_0 t + A\cos\omega t$

C. $\upsilon_0 \sin\omega_0 t + \frac{A}{2}\sin(\omega_0 - \omega)t + \frac{A}{2}\sin(\omega_0 + \omega)t$

D. $\upsilon_0 \sin[\omega_0 (1 + 0.01 A\sin\omega t)t]$

Ans. C.

Sol:

Given, modulating signal,

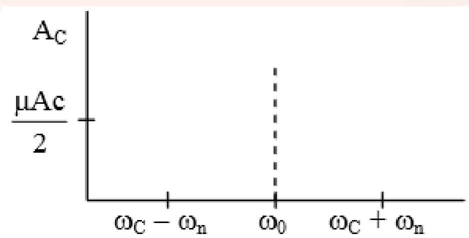
$A_m = A \cos \omega t$

Carrier wave, $A_c = v_0 \sin \omega_0$

In amplitude modulation, modulated wave is given by

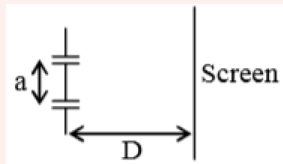
According to standard result is should be

$= A_C \sin\omega_C t + \frac{\mu A_C}{2}\sin(\omega_0 - \omega)t + \frac{\mu A}{2}\sin(\omega_0 + \omega)t$



So, Answer C

11. The figure shows a Young's double slit experimental setup. It is observed that when a thin transparent sheet of thickness t and refractive index μ is put in front of one of the slits, the central maximum gets shifted by a distance equal to n fringe widths. If the wavelength of light used it λ, t will be:



A. $\frac{nD\lambda}{a(\mu-1)}$ B. $\frac{2D\lambda}{a(\mu-1)}$

C. $\frac{D\lambda}{a(\mu-1)}$ D. $\frac{2nD\lambda}{a(\mu-1)}$

Ans. (Bonus)

Sol:

Path difference introduced by a slab of thickness t and refractive index μ is given

$\Delta = (\mu - 1)t$

Position of the fringe is $x = \frac{\Delta D}{d} = \frac{(\mu-1)tD}{d}$

Also, fringe width is given by

$x = \frac{\Delta D}{d} = \frac{(\mu-1)tD}{d}$

Path difference $\Delta = (\mu - 1)t$

$$(\mu-1)t\frac{D}{d} = \frac{n\lambda D}{d}$$

$$f = \frac{n\lambda}{\mu-1}$$

12. The total number of turns and cross-section area in a solenoid is fixed. However, its length L is varied by adjust the separation between windings. The inductance of solenoid will be proportional to:

A. $1/L^2$ B. L

C. $1/L$ D. L^2

Ans. C

Sol:

Self-Inductance of solenoid =

$$\phi = NBA = L^{'}I$$

$$N\mu_0 nI\pi r^2 = L^{'}I$$

$$N\mu_0 \frac{N}{L}\pi r^2 I = L^{'}I$$

$$L^{'} \propto \frac{1}{L}$$

13. The electric field of light wave is given as

$$\vec{E} = 10^{-3}\cos\left(\frac{2\pi x}{5\times10^{-7}} - 2\pi\times6\times10^{14}t\right)x\frac{N}{C}$$

This light falls on a metal plate of work function 2eV. The stopping potential of the photo-electrons is:

Given, $E\,(\text{in eV}) = \frac{12375}{\lambda\,(\text{in Å})}$

A. 0.72 V B. 0.48 V

C. 2.48 V D. 2.0 V

Ans. B

Sol:

Given

$$E = 10^{-3}\cos\left(\frac{2\pi x}{5\times10^{-7}} - 2\pi\times6\times10^{14}t\right)\hat{x}\,NC^{-1}$$

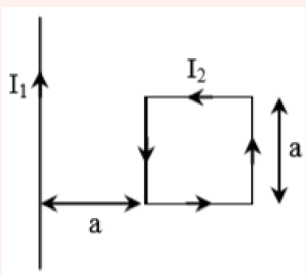
By comparing it with the general equation of electric field of light, i.e.,

$E = E_0\cos(kx - \omega t)\,\hat{x},$ we get

$k = \frac{2\pi}{5\times10^{-7}} = 2\pi/\lambda$ (from definition $k = 2\pi/\lambda$)

$\Rightarrow \lambda = 5\times10^{-7}\,m = 5000\,\overset{o}{A}$

14. A rigid square loop of side 'a' and carrying current I_2 is lying on a horizontal surface near a long current I_1 carrying wire in the same plane as shown in figure. The net force on the loop due to the wire will be:



A. Attractive and equal to $\frac{\mu_o I_1 I_2}{3\pi}$

B. Repulsive and equal to $\frac{\mu_o I_1 I_2}{2\pi}$

C. Repulsive and equal to $\frac{\mu_o I_1 I_2}{4\pi}$

D. Zero

Ans. C.

Sol:

Force on a wire 1 in which current I_1 is flowing due to another wire 2 which are separated by a distance r is given as

$\mathbf{F} = I_1\,(l \times \mathbf{B}_2)$

Or $F = \frac{\mu_0 I_1 I_2}{2\pi r} . l \sin\theta \quad \left[\because \mathbf{B}_2 = \frac{\mu_0 I_2}{2\pi r}\right]$

Force of PQ

$F_1 = I_2 B_1 A$

$= I_2 \frac{\mu_0 I_1}{2\pi a} a$

Force of RS

$F_2 = I_2 B_2 a$

$= I_2 \frac{\mu_0 I_1}{2\pi(2a)} a$

Net force $F_1 - F_2$

$= \frac{\mu_0 I_1 I_2}{4\pi}$ (Repulsive)

15. A simple pendulum oscillating in air has period T. The bob of the pendulum is completely immersed in a non-viscous liquid. The density of the liquid is $\frac{1}{16}$th of the material of the bob. If the bob is inside liquid all the time, its period of oscillation is this liquid is:

A. $2T\sqrt{\frac{1}{14}}$ B. $2T\sqrt{\frac{1}{10}}$

C. $4T\sqrt{\frac{1}{15}}$ D. $4T\sqrt{\frac{1}{14}}$

Ans. C

Sol:

We know that

Time period of a pendulum is given by

$T = 2\pi\sqrt{L / g'_{eff}}$

Given that

$T = 2\pi$

When immersed in liquid

Tension $\Rightarrow T' = mg - m'g$

$$T' = mg\left(1 - \frac{m'}{m}\right)$$

$$T' = T\left(1 - \frac{Vd\ell}{Vd_0}\right)$$

$$T' = T\left(1 - \frac{1}{16}\right)$$

$$T' = \frac{15T}{16}$$

$$g' = \frac{15}{16}g$$

$$\frac{T'}{T} = \sqrt{\frac{16}{15}}$$

$$T' = \frac{4T}{\sqrt{5}}$$

16. A concave mirror for face viewing has focal length of 0.4 m. The distance at which you hold the mirror from your face in order to see your image upright with a magnification of 5 is:

A. 0.16 m B. 1.60 m

C. 0.32 m D. 0.24 m

Ans. C

Sol:

Magnification produced by a mirror can also be give as

$$m = \frac{f}{f - u}$$

Substituting the given values, we get

$$5 = \frac{-0.4}{-0.4 - u}$$

or u = -0.32 m

17. A moving coil galvanometer has resistance 50 Ω and it indicates full deflection at 4 mA current. A voltmeter is made using this galvanometer and a 5 kΩ resistance. The maximum voltage, that can be measured using this voltmeter, will be close to:

A. 10 V B. 15 V

C. 40 V D. 20 V

Ans. D.

Sol:

Given, resistance of galvanometer, $R_g = 50\ \Omega$

Current, $I_g = 4mA = 4 \times 10^{-3}$ A

Resistance used in converting a galvanometer in voltmeter,

$R = 5\ k\Omega = 5000\ \Omega$

The maximum voltage is given by

$V = I_g (G + S)$

$V = 4 \times 10^{-3} (50 + 5000)$

$= 4 \times 10^{-3} \times 5050$

$= 20V$ (approximate)

18. A capacitor with capacitance 5 μF is charged to 5 μC. If the plates are pulled apart to reduce the capacitance to 2 μF, how much work is done?

A. 2.5×10^{-6} J B. 3.75×10^{-6} J

C. 2.16×10^{-6} J D. 6.25×10^{-6} J

Ans. B

Sol:

Potential energy stored in a capacitor is

$$U = \frac{1}{2}qV = \frac{1q^2}{2C}$$

So, initial energy of the capacitor, $U_i = \frac{1}{2}q^2 / C_1$

Final energy of the capacitor, $U_f = \frac{1}{2}q^2 / C_2$

Work done = $U_f - U_i$

$$= \frac{q^2}{2C_2} - \frac{q^2}{2C_1}$$

$$= \frac{\left(5\times10^{-6}\right)}{2}\left(\frac{1}{2\times10^{-6}} - \frac{1}{5\times10^{-6}}\right)$$

$$= \frac{15}{4}\times10^{-6}$$

$$= 3.75\times10^{-6}$$

19. A solid sphere of mass 'M' and radius 'a' is surrounded by a uniform concentric spherical shell of thickness 2a and mass 2M. The gravitational field at distance '3a' from the centre will be:

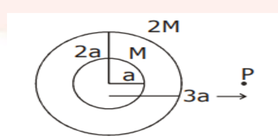
A. $\frac{2GM}{3a^2}$ B. $\frac{GM}{9a^2}$

C. $\frac{2GM}{9a^2}$ D. $\frac{GM}{3a^2}$

Ans. D

Sol:

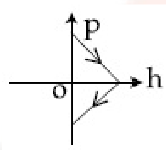
According to the question

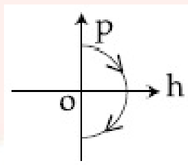


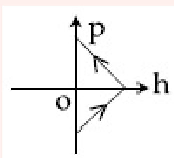
$$g_p = \frac{GM}{(3a)^2} + \frac{G2M}{(3a)^2}$$

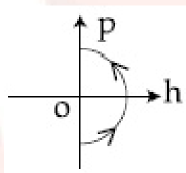
$$g_p = \frac{3GM}{9a^2} = \frac{GM}{3a^2}$$

20. A ball is thrown vertically up (taken as + z-axis) from the ground. The correct momentum-height (p-h) diagram is:

A. 

B. 

C. 

D. 

Ans. B

Sol:

When a ball is thrown vertically upward, then the acceleration of the ball,

a = acceleration due to gravity (g) (acting I the downward direction).

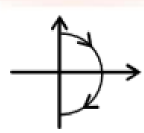
Now, using the equation of motion.

When thrown upward, (h = 0) v → max

$v^2 = u^2 - 2as$

$v^2 = u^2 - 2gh$ (Parabola)

When downward (h → max, V → 0)



21. An NPN transistor is used in common emitter configuration as an amplifier with 1 kΩ load resistance. Signal voltage of 10 mV is applied across the base-emitter. This produces a 3 mA change in the collector current and 15 μA change in the base current of the amplifier. The input resistance and voltage gain are:

A. 0.67 kΩ, 200 B. 0.33 kΩ, 1.5

C. 0.67 kΩ, 300 D. 0.33 kΩ, 300

Ans. C

Sol:

Given, load resistance, $R_L = 1\ k\Omega$

Current input = 15×10^{-6}

Current output = 3×10^{-3}

$R_0 = 1000$

$V_{in} = r_{in} \times I_{in}$

$$r_{in} = \frac{2000}{3} = .67\ k\Omega$$

$$\text{Voltage gain} = \frac{V_0}{V_i} = \frac{1000 \times 3 \times 10^{-3}}{10 \times 10^{-3}} = 300$$

22. The magnetic field of a plane electromagnetic wave is given by:

$B = B_0\,\hat{i}\left[\cos(kz - \omega t)\right] + B_1\,\hat{j}\cos(kz + \omega t)$ where $B_0 = 3 \times 10^{-5}\ T$ an $B_1 = 2 \times 10^{-6}\ T$. The rms value of the force experienced by a stationary charge $Q = 10^{-4}\ C$ at z = 0 is closest to:

A. 3×10^{-2} N B. 0.9 N

C. 0.6 N D. 0.1 N

Ans. A

Sol:

Given, magnetic field of an electromagnetic wave is

$B = B_0\left[\cos\left(kz-\omega t\right)\right]\hat{i} + B_1\left[\cos(kz+\omega t)\right]\hat{j}$

Here, $B_0 = 3 \times 10^{-5}$ T and $B_1 = 2 \times 10^{-6}$ T

Also, stationary charge, $Q = 10^{-4}$ C at z = 0

As charge is released from rest at z = 0, in this condition.

Maximum Electric Field = BC

$$F_{net} = q\vec{E}_{net}$$

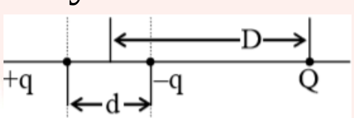
$$= qC\left(-3\times10^{-5}\hat{j} - 2\times10^{-6}\hat{i}\right)$$

$$= 10^{-4}\times3\times10^{8}\sqrt{\left(3\times10^{-5}\right)^2 + \left(2\times10^{-6}\right)^2}$$

$$= .9 \text{ N}$$

$$F_{rms} = \frac{F_0}{\sqrt{2}} = .6 \text{ N}$$

23. A system of three charges are placed as shown in the figure:



If D>>d, the potential energy of the system is best given by:

A. $\frac{1}{4\pi\in_o}\left[-\frac{q^2}{d} - \frac{qQd}{D^2}\right]$

B. $\frac{1}{4\pi\in_o}\left[-\frac{q^2}{d} + \frac{2qQd}{D^2}\right]$

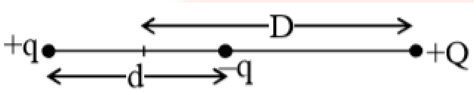
C. $\frac{1}{4\pi\in_o}\left[+\frac{q^2}{d} + \frac{qQd}{D^2}\right]$

D. $\frac{1}{4\pi\in_o}\left[-\frac{q^2}{d} - \frac{qQd}{2D^2}\right]$

Ans. A

Sol:

The system of two charges, i.e., +q and -q that are separated by distance d can be considered as a dipole. Thus, the charge Q would be at D distance from the centre of an electric dipole on its axis line.
So, the total potential energy of the system will be due to two components.



the potential energy of the system is given by $u = -\frac{kq^2}{d} - \frac{kQ(pd)}{D^2}$

$$= -k\left[\frac{q^2}{d} + \frac{qQd}{D^2}\right]$$

24. For a given gas at 1 atm pressure, rms speed of the molecules is 200 m/s at 127°C. At 2 atm pressure and at 227°C, the rms speed of the molecules will be:

A. 100 m/s B. $100\sqrt{5}$ m/s

C. 800 m/s D. $80\sqrt{5}$ m/s

Ans. B

Sol: The rms speed of the molecules

$$V_{rms} = \sqrt{\frac{3RT}{M_w}}$$

$$\frac{V_2}{V_1} = \sqrt{\frac{T_2}{T_1}}$$

$$\frac{V_2}{200} = \sqrt{\frac{500}{400}}$$

$$V_2 = 100\sqrt{5}$$

25. The pressure wave,

$P = 0.01 \sin [1000t - 3x] \, Nm^{-2}$, corresponds to the sound produced by a vibrating blade on a day when atmospheric temperature is 0°C. On some other day when temperature is T, the speed of sound produced by the same blade and at the same frequency is found to be $336 \, ms^{-1}$. Approximate value of T is:

A. 4°C B. 11°C

C. 12°C D. 15°C

Ans. A

Sol:

Given, $p = 0.01 \sin (1000t - 3x) \, N/m^2$

Comparing with the general equation of pressure wave of sound i.e., $p_0 \sin (\omega t - kx)$.

$$\text{Speed of sound} = \frac{w}{k}$$

$$= \frac{1000}{3}$$

$$v \propto \sqrt{T}$$

$$\frac{V_2}{V_1} = \sqrt{\frac{T_2}{T_1}}$$

$$\frac{336}{1000/3} = \sqrt{\frac{7}{273}}$$

$$T = 277 \, K \, (\text{appr.})$$

$$T = 4°C$$

26. In the density measurement of a cube, the mass and edge length are measured as (10.00 ± 0.10) kg and (0.10 ± 0.01) m, respectively. The error in the measurement of density is:

A. 0.01 kg/m^3 B. 0.31 kg/m^3

C. 0.10 kg/m^3 D. 0.07 kg/m^3

Ans. (Bonus)

Sol: It is given that

$M = 10 \pm 0.10 \quad \ldots(i)$

$\ell = 0.10 \pm 0.01 \quad \ldots(ii)$

$$\rho = \frac{M}{\ell^3} = \frac{10}{(0.1)^3} = 10^4$$

∵ Permissible error in density is

$$\frac{\Delta\rho}{\rho} = \frac{\Delta M}{M} + \frac{3\Delta\ell}{\ell}$$

Substituting the value from equation (i) and (ii)

$$= \frac{0.10}{10} + \frac{3 \times .01}{0.10}$$

$$= \frac{1}{100} + \frac{3}{10}$$

$$\frac{\Delta\rho}{\rho} = \frac{31}{100}$$

$$\frac{\Delta\rho}{\rho} = 0.31$$

This Ans. is not for error in density it is relative error. So (Q) should be Bonus.

27. The stream of a river is flowing with a speed of 2 km/h. A swimmer can swim at a speed of 4 km/h. What should be the direction of the swimmer with respect to the flow of the river to cross the river straight?

A. 60°   B. 150°

C. 90°   D. 120°

Ans. D

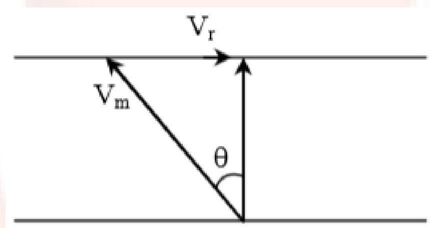
Sol:

Let the velocity of the swimmer is

$v_s = 4km/h$

and velocity of river is $v_r = 2km/h$

Also, angle of swimmer with the flow of the river (down stream) is α as shown in the figure below



$$V_m = \sin\theta = V_r$$

$$4\sin\theta = 2$$

$$\sin\theta = \frac{1}{2}$$

$$\theta = 30°$$

Angle from river flow = $30^o + 90^o = 120^o$

28. A string is clamped at both the ends and it is vibrating in its 4^{th} harmonic. The equation of the stationary wave is Y = 0.3 sin(0.157*x*) cos(200πt). The length of the string is: (All quantities are in SI units.)

A. 80 m   B. 40 m

C. 60 m   D. 20 m

Ans. A

Sol:

Given equation of stationary wave is

Y = 0.3 sin (0.157x) cos (200 πt)

Comparing it with general equation of stationary wave i.e., Y = a sin kx cos ωt, we get

$$\frac{2\pi}{\lambda} = .157$$

$$\lambda = 40$$

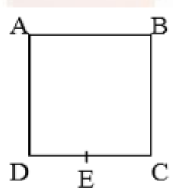
4^{th} harmonic real

$$\ell = 4\frac{\lambda}{2}$$

$$\ell = 2\lambda$$

$$= 80 \text{ m}$$

29. A wire of resistance R is bent to form a square ABCD as shown in the figure. The effective resistance between E and C is: (E is mid-point of arm CD)



A. $\frac{3}{4}$R B. $\frac{1}{16}$R

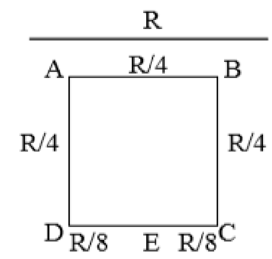
C. $\frac{7}{64}$R D. R

Ans. C

Sol:

Let the length of each side of square ABCD is a.

∴ Resistance per unit length of each side R/4.



$$R_{EDABC} = \frac{R}{4} + \frac{R}{4} + \frac{R}{4} + \frac{R}{8} = \frac{7R}{8}$$

$$R_{EC} = \frac{R}{8}$$

$$\text{Effective resistance} = \frac{\frac{R}{8} \times \frac{7R}{8}}{\frac{R}{8} + \frac{7R}{8}} = \frac{7R}{64}$$

30. The following bodies are made to roll up (without slipping) the same inclined plane from a horizontal plane: (i) a ring of radius R, (ii) a solid cylinder of radius $\frac{R}{2}$ and (iii) a solid sphere of radius $\frac{R}{4}$. If, in each case, the speed of the center of mass at the bottom of the incline is same, the ratio of the maximum heights they climb is:

A. 2: 3: 4 B. 14: 15: 20

C. 10: 15: 7 D. 4: 3: 2

Ans. (Bonus)

Sol:

From questions,

let height attained by ring = h_1

Height attained by cylinder = h_2

Height attained by sphere = h_3

As we know that for a body which is rolling up an inclined plane (without slipping), follows the law of conservation of energy.

By energy conservation

$$\frac{1}{2}mv^2\left(1+\frac{k^2}{r^2}\right)=mgh$$

$$h \propto \left(1+\frac{k^2}{r^2}\right)$$

$$h_1 : h_2 : h_3 \rightarrow (1+1) : \left(1+\frac{1}{2}\right) : \left(1+\frac{2}{5}\right)$$

$$\Rightarrow \quad 2 : \frac{3}{2} : \frac{7}{5}$$

$$\Rightarrow \quad 20 : 15 : 14$$

So, Ans Should be 20 : 15 : 14

Chemistry

1. Which of the following statements is not true about sucrose?

A. On hydrolysis, it produces glucose and fructose

B. The glycosidic linkage is present between C_1 of α-glucose and C_1 of β-fructose

C. It is a non-reducing sugar

D. It is also named as invert sugar

Ans. B

Sol:

Statement (B) is not true for sucrose. It is linked through a glycoside linkage between C-1 of

α- glucose and C-2 of β-fructose.

Sucrose $\xrightarrow{+H_2O}$ α-D-Glucose
+
β-D-Fructose

In β-D-Fructose → C-2 in involved

In β-D-Glucose → C-1 in involved

2. The element having greatest difference between its first and second ionization energies, is:

A. Ba B. K

C. Ca D. Sc

Ans. B

Sol:

The electronic configuration of given elements are as follows:

K(19) = $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1$

Mg(12) = $1s^2\ 2s^2\ 2p^6\ 3s^2$

Sr(38) = $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^6\ 5s^2$

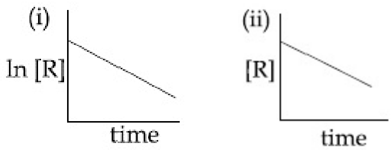
Sc(21) = $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^1$

First ionization enthalpy (I.E.) of K is lowest among the given options.

Second ionization enthalpy of K is highest among the given options.

It can be concluded that ($I.E_2 - I.E_1$) value is maximum for K (potassium).

3. The given plots represent the variation of the concentration of a reactant R with time for two different reaction (i) and (ii). The respective orders of the reactions are:



A. 0, 1 B. 1, 0

C. 0, 2 D. 1, 1

Ans. B

Sol:

In first order reaction, the rate expression depends on the concentration of one species only having power equal to unity.

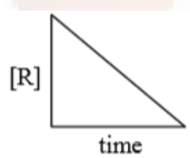
nr → products

$$\frac{d[r]}{dt} = k[r]$$

For zero order reaction

$$(a - x) = -K_0 t + a$$

$$[A]_t = -K_0 t + [A]_0$$



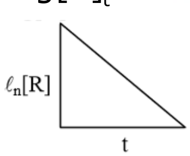
for I order reaction

$$K_1 = \frac{2.3}{t} \log \frac{a}{a - x}$$

$$K_1 = \frac{1}{t} \ln \frac{a}{a - x}$$

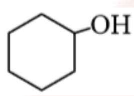
$$K_1 t = \log a - \log a - x$$

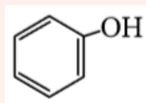
$$\log[A]_t = -K_1 t + \log[A_0]$$

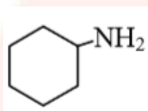


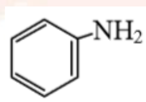
4. The organic compound that gives following qualitative analysis is:

Test	Inference
a. Dil. HCl	Insoluble
b. NaOH solution	soluble
c. Br_2/water	Decolourization

A. 

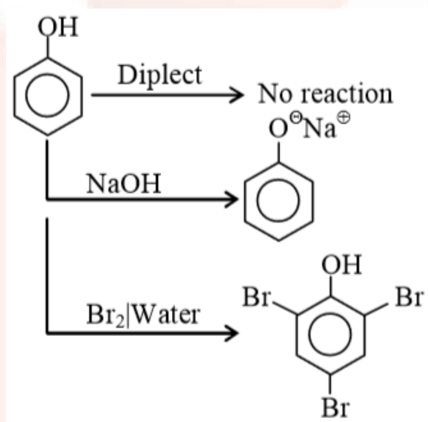
B. 

C. 

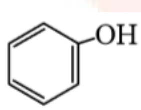
D. 

Ans. B

Sol:



So, from the above figure we get that the organic compound is



5. The degenerate orbitals of $[Cr(H_2O)_6]^{3+}$ are:

A. d_{xz} and d_{yz} B. $d_{x^2-y^2}$ and d_{xy}

C. d_{z^2} and d_{xz} D. d_{yz} and d_{z^2}

Ans. A

Sol:

The degenerate orbitals of $[Cr(H_2O)_6]^{3+}$ are d_{xz} and d_{yz}.

Electronic configuration of Cr^{3+} is $3d^5\ 4s^1$. The five d-orbitals in an isolated gaseous atom or ion have same energy, i.e., they are degenerate.

6. Match the catalysts (**Column I**) with product (**Column II**).

Column I	**Column II**
Catalyst	**Product**
1. V_2O_5	i. Polyethylene
2. $TiCl_4/Al(Me)_3$	ii. Ethanol
3. $PdCl_2$	iii. H_2SO_4
4. Iron Oxide	iv. NH_3

A. 1-ii; 2-iii; 3-i; 4-iv

B. 1-iii; 2-iv; 3-i; 4-ii

C. 1-iv; 2-iii; 3-ii; 4-i

D. 1-iii; 2-I; 3-ii; 4-iv

Ans. D.

Column I	**Column II**
Catalyst	**Product**

1. V_2O_5	iii.	H_2SO_4
2. $TiCl_4/Al(Me)_3$	i.	Polyethylene
3. $PdCl_2$	ii.	Ethanol
4. Iron Oxide	iv.	NH_3

7. The osmotic pressure of a dilute solution of an ionic compound XY in water is four times that of a solution of 0.01 M $BaCl_2$ in water. Assuming complete dissociation of the given ionic compounds in water, the concentration of XY (in mol L^{-1}) in solution is:

A. 4×10^{-4} B. 6×10^{-2}

C. 16×10^{-4} D. 4×10^{-2}

Ans. B

Sol: Osmotic pressure is proportional to the molarity (C) of the solution at a given temperature $\pi = CRT$

Concentration of $BaCl_2$ = 0.01 M.

$$\pi_{xy} = 4\pi_{BaCl_2}$$
$$iCRT = 4(iCRT)$$
$$2C = 4 \times 3(0.01)$$
$$C = 0.06 = 6 \times 10^{-2}\ M$$

8. Among the following, the set of parameters that represents path functions, is:

1. q + w 2. Q

3. w 4. H – TS

A. 1 and 2 B. 2, 3 and 4

C. 2 and 4 D. 1, 2 and 3

Ans. A

Sol:

(A) q + w = ΔE state function

(B) q = Path function

(C) w = Path function

(D) H – TS = G = State function

9. The aerosol is a kind of colloid in which:

A. gas is dispersed in liquid
B. solid is dispersed in gas
C. liquid is dispersed in water
D. gas is dispersed in solid

Ans. B

Sol:

The aerosol is a kind of colloid in which solid is dispersed in gas. e.g., smoke, dust.

10. The major product of the following reaction is:

$$CH_3CH = CHCO_2CH_3 \xrightarrow{LiAlH_4}$$

A. $CH_3CH_2CH_2CO_2CH_3$
B. $CH_3CH_2CH_2CHO$
C. $CH_3CH{=}CHCH_2OH$
D. $CH_3CH_2CH_2CH_2OH$

Ans. C

Sol:

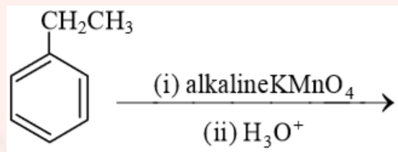
$$CH_3\text{–}CH = CH - \underset{\underset{O}{\|}}{C} - O - CH_3$$

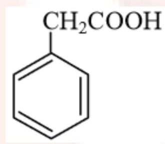
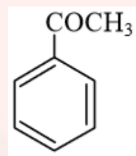
$$\downarrow LiAlH_4 \quad \text{(Reduction)}$$

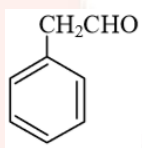
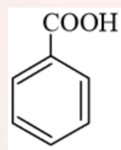
$$CH_3\text{–}CH = CH - CH_2 - OH$$

(1°-Alcohol)

11. The major product of the following reaction is:

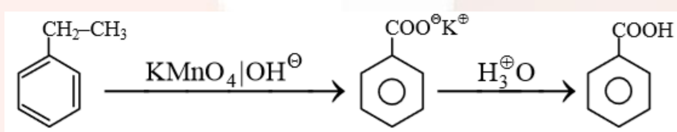


A.  B. 

C.  D. 

Ans. D

Sol:



12. Among the following, the molecule expected to be stabilized by anion formation is:

C_2, O_2, NO, F_2

A. F_2 B. O_2

C. NO D. C_2

Ans. D

Sol: The molecule in which B.O. ↑ with formation of anion, will get stabilize.

	F_2 ——	F_2^-		C_2 →	C_2^{-1}
BO	1	0.5	BO	2	2.5
	NO ——	NO^-		O_2 →	O_2^{-1}
BO	2.5	2.0		2	1.5

The value fo bond order of C_2^- is highest among the given options.
The bond length decreases as bond order increases. As a result, stabilty of a moelcule increases.

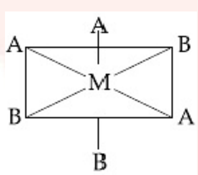
13. Excessive release of CO_2 into the atmosphere results in:

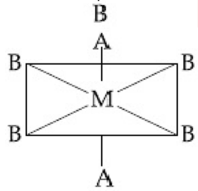
A. depletion of ozone
B. formation of smog
C. mmmglobal warming
D. polar vortex

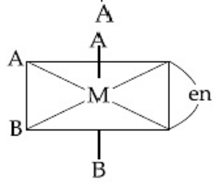
Ans. C

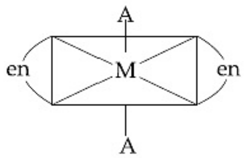
Sol: CO_2 causes global warming.

14. The one that will show optical activity is:
(en = ethane-1, 2-diamine)

A. 

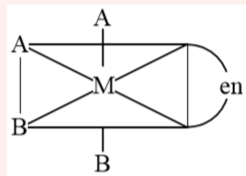
B. 

C. 

D. 

Ans. C

Sol:



has no plane of symmetry
or centre of symmetry, hence it is optically active.

15. C_{60}, an allotrope of carbon contains:

A. 12 hexagons and 20 pentagons.
B. 16 hexagons and 16 pentagons.
C. 20 hexagons and 12 pentagons.
D. 18 hexagons and 14 pentagons.

Ans. C

Sol:

$C_{60}$is aromatic allotrope of carbon containing 12 pentagons and 20 hexagons.

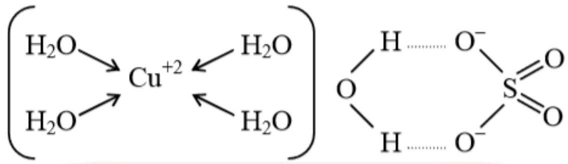
16. The number of water molecule(s) not coordinated to copper ion directly in $CuSO_4 \cdot 5H_2O$, is:

A. 3 B. 4
C. 5 D. 1

Ans. D

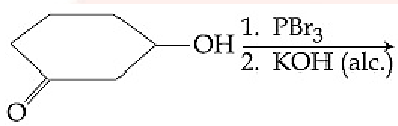
Sol:

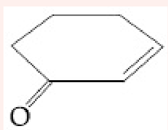
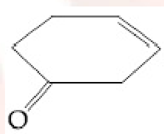
In $CuSO_4.5H_2O$, one molecule of water is indirectly connected to Cu.

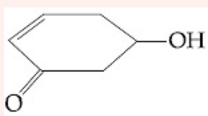
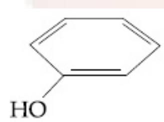


∴ 4, H_2O molecules are coordinated directly.

17. The major product of the following reaction is:

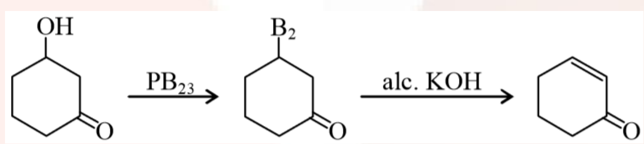


A.  B. 

C.  D. 

Ans. A

Sol:



18. Consider the van der Waals constants, a and b, for the following gases.

Gas	Ar	Ne	Kr	Xe
a/(atm dm^6 mol^{-2})	1.3	0.2	5.1	4.1
b/(10^{-2} dm^3 mol^{-2})	3.2	1.7	1.0	5.0

Which gas is expected to have the highest critical temperature?

A. Ne　　B. Xe

C. Kr　　D. Ar

Ans. C

Sol:

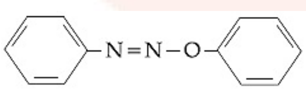
Critical temperature is the temperature of a gas above which it cannot be liquefied what ever high the pressure may be. The kinetic energy gas molecules above this temperature is sufficient energy to overcome the attractive forces. It is represented as T_c. Critical temperature is given by

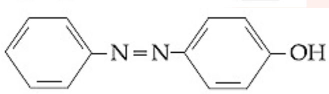
$$T_c = \frac{8a}{27Rb}$$

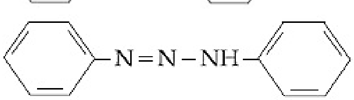
$$T_c \propto \frac{a}{b}$$

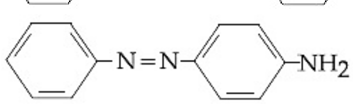
Kr has highest ratio or $\frac{a}{b}$

19. Aniline dissolved in dilute HCl is reacted with sodium nitrite at 0°C. This solution was added dropwise to a solution containing equimolar mixture of aniline and phenol in dil. HCl. The structure of the major product is:

A. 

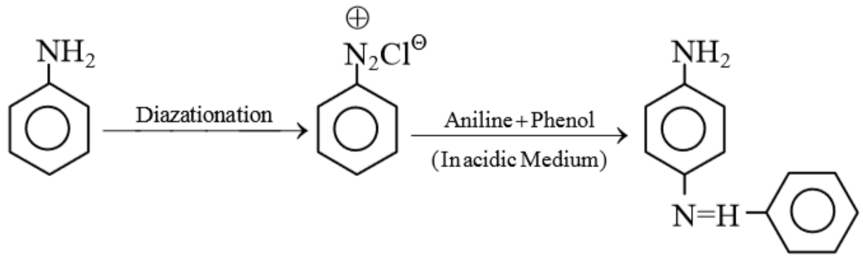
B. 

C. 

D. 

Ans. D.

Sol:



20. The correct order of the oxidation states of nitrogen in NO, N_2O, NO_2 and N_2O_3 is:

A. $N_2O < N_2O_3 < NO < NO_2$
B. $NO_2 < NO < N_2O_3 < N_2O$
C. $N_2O < NO < N_2O_3 < NO_2$
D. $NO_2 < N_2O_3 < NO < N_2O$

Ans. C.

Sol:

The correct increasing order of oxidation state of nitrogen for nitrogen oxides is

$$\overset{+1}{N_2}O < \overset{+2}{N}O < \overset{+3}{N_2}O_3 < \overset{+4}{N}O_2$$

21. For any given series of spectral lines of atomic hydrogen, let $\Delta\bar{v} = \bar{v}_{max} - \bar{v}_{min}$ be the difference in maximum and minimum frequencies in cm^{-1}. The ratio $\Delta\bar{v}_{Lyman} / \Delta\bar{v}_{Balmer}$ is:

A. 5: 4 B. 9: 4
C. 27: 5 D. 4: 1

Ans. B.

Sol:

$$\frac{\Delta\bar{v}_{Lyman}}{\Delta\bar{v}_{Balmar}} = \frac{\Delta\bar{v}_{max} - \Delta\bar{v}_{min}}{\Delta\bar{v}_{max} - \Delta\bar{v}_{min}} \qquad \text{for Hydrogen } \bar{v} = RZ^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

$$= \frac{\left[\frac{1}{1} - \frac{1}{\infty}\right] - \left[\frac{1}{1} - \frac{1}{4}\right]}{\left[\frac{1}{4} - \frac{1}{\infty}\right] - \left[\frac{1}{4} - \frac{1}{9}\right]}$$

$$= \frac{\frac{1}{4}}{\frac{1}{9}} = \frac{9}{4}$$

$$= 9 : 4$$

22. The ore that contains the metal in the form of fluoride is:

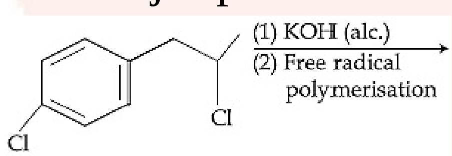
A. sphalerite B. malachite

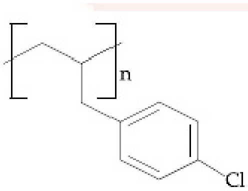
C. cryolite D. magnetite

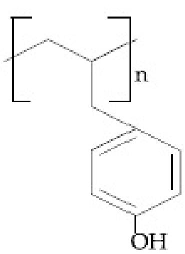
Ans. C

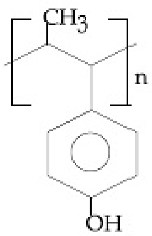
Sol: cryolite = Na_3AlF_6 is the ore that contains the metal in the form of fluoride.

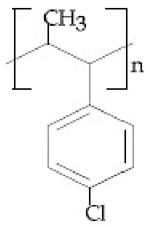
23. The major product of the following reactions is:



A. 

B. 

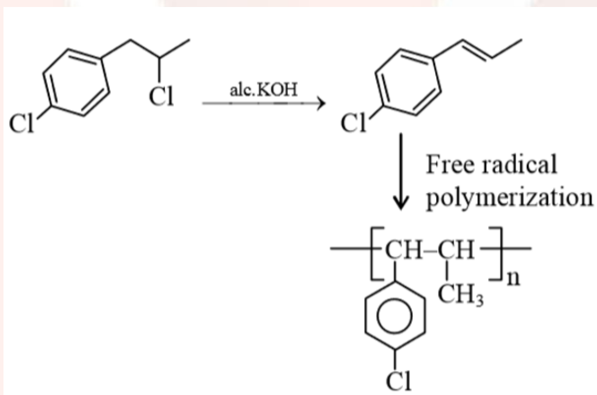
C. 

D. 

Ans. D

Sol:

In presence of alc, KOH, the given halide undergoes elimination reaction.



24. The standard Gibbs energy for the given cell reaction in kJ mol^{-1} at 298 K is:

$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$,

$E^0 = 2$ V at 298 K

(Faraday's constant, F = 96000 C mol^{-1})

A. –192 B. 384

C. 192 D. –384

Ans. D.

Sol:

Gibbs energy of the reaction is related to E^o_{cell} by the following formula.

$$\Delta G^o = -nFE^o_{cell}$$

ΔG^o = Gibbs energy of cell

nF = amount of charge passed

E = EMF of a cell

The standard Gibbs energy

$$\Delta G = -nFE_{cell}$$
$$= -2(96000)^2$$
$$= -384000 \text{ J/mole}$$
$$= -384 \text{ KJ/mole}$$

25. The major product of the following reaction is:

$$CH_3C \equiv CH \xrightarrow[\text{(ii) DI}]{\text{(i) DCl (1 equiv.)}}$$

A. $CH_3C(I)(CI)CHD_2$

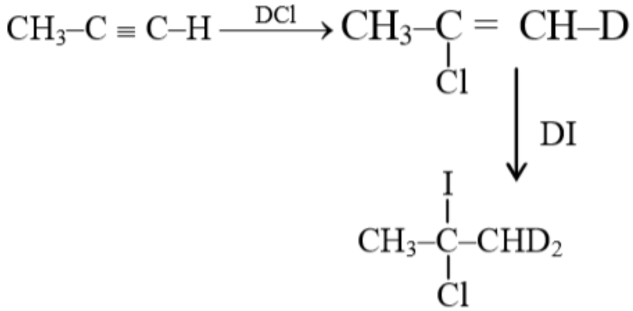
B. $CH_3CD(I)CHD(Cl)$

C. $CH_3CD_2CH(Cl)(I)$

D. $CH_3CD(Cl)CHD(I)$

Ans. A

Sol:



26. Magnesium powder burns in air to give:

A. MgO and $Mg(NO_3)_2$
B. $Mg(NO_3)_2$ and Mg_3N_2
C. MgO only
D. MgO and Mg_3N_2

Ans. D.

Sol:

Magnesium powder burns in air to given MgO and Mg_3N_2. MgO does not combine with excess oxygen to give any superoxide. Mg reacts with nitrogen to form magnesium nitride (Mg_3N_2).

Air contain both N_2 and O_2

$$Mg + O_2 \xrightarrow{\Delta} MgO$$

$$Mg + N_2 \xrightarrow{\Delta} Mg_3N_2$$

$\therefore$ Both MgO and Mg_3N_2 are formed.

27. For a reaction,

$N_2(g) + 3H_2(g) \rightarrow 2\,NH_3(g)$; identify dihydrogen ($H_2$) as a limiting reagent in the following reaction mixtures.

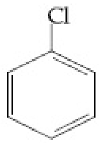
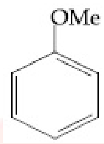
A. 35 g N_2 + 8 g of H_2
B. 14 g N_2 + 4 g of H_2
C. 56 g N_2 + 10 g of H_2
D. 28 g N_2 + 6 g of H_2

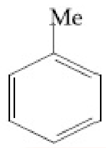
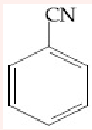
Ans. C.

Sol:

The reactant which is present in the lesser amount, i.e., which limits the amount of product formed is called limiting reagent. When 56g of N_2 + 10g is taken as a combination then dihydrogen (H_2) act as a limiting reagent in the reaction.

28. **The increasing order of reactivity of the following compounds towards aromatic electrophilic substitution reaction is:**

1.  2. 

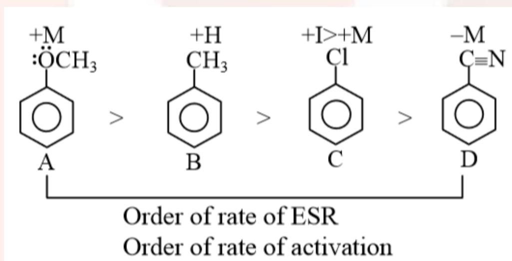
3.  4. 

A. 2 < 3 < 1 < 4 B. 4 < 1 < 3 < 2

C. 4 < 2 < 1 < 3 D. 1 < 2 < 3 < 4

Ans. B

Sol:



29. Liquid 'M' and liquid 'N' form an ideal solution. The vapour pressures of pure liquids 'M' and 'N' are 450 and 700 mmHg, respectively, at the same temperature. Then correct statement is:

(x_M = Mole fraction of 'M' in solution;
x_N = Mole fraction of 'N' in solution;
y_M = Mole fraction of 'M' in vapour phase;
y_N = Mole fraction of 'N' in vapour phase)

A. $\frac{x_M}{x_N} = \frac{y_M}{y_N}$

B. $\frac{x_M}{x_N} > \frac{y_M}{y_N}$

C. $\frac{x_M}{x_N} < \frac{y_M}{y_N}$

D. $(x_M - y_M) < (x_N - y_N)$

Ans. B

Sol:

The solution of volatile liquids the partial vapour pressure of each component of the solution is directly proportional to its mole fraction present in solution. This is know as Raoul's law. Liquid M and N form an ideal solution. f

Given that

The vapour pressures of pure liquids'M' and 'N' are 450 and 700 mmHg, respectively

$P^\circ{}_M = 450$

$P^\circ{}_N = 700$

$y_M = \frac{P^\circ{}_M X_M}{P_S}$

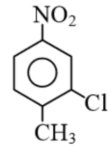
$y_N = \frac{P^\circ{}_N X_N}{P_S}$

$\frac{y_M}{y_N} = \frac{P^\circ{}_M}{P^\circ{}_N} \frac{X_M}{X_N}$

$\frac{y_M}{y_N} = \frac{450}{700} \frac{X_M}{X_N}$

$\frac{X_M}{X_N} > \frac{y_M}{y_N}$

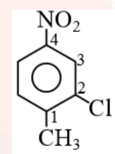
30. The correct IUPAC name of the following compound is:



A. 5-chloro-4-methyl-1-nitrobenzene
B. 3-chloro-4-methyl-1-nitrobenzene
C. 2-chloro-1-methyl-4-nitrobenzene
D. 2-methyl-5-nitro-1-chlorobenzene

Ans. B

Sol:

The correct IUPAC name of the following compound  is:

2-Chloro-1-methyl-4-nitrobenzene

Mathematics

1. Let S be the set of all values of *x* for which the tangent to the curve $y = f(x) = x^3 - x^2 - 2x$ at (*x*, *y*) is parallel to the line segment joining the points (1, *f*(1)) and (–1, *f*(–1)), then S is equal to:

A. $\left\{\frac{1}{3}, -1\right\}$ B. $\left\{-\frac{1}{3}, 1\right\}$

C. $\left\{\frac{1}{3}, 1\right\}$ D. $\left\{-\frac{1}{3}, -1\right\}$

Ans. B.

Sol: Given that

$y = f(x) = x^3 - x^2 - 2x$

slope of tangent $\frac{dy}{dx} = f'(x) = 3x^2 - 2x - 2$

This tangent is parallel to line segment joining points (1, f(1)) and (–1, f(–1)) $\therefore m_1 = m_2$

$\Rightarrow \quad 3x^2 - 2x - 2 = \frac{f(-1) - f(1)}{-1 - 1}$

$\Rightarrow \quad 3x^2 - 2x - 2\,\frac{(-1 - 1 + 2) - (1 - 1 - 2)}{-2}$

$\Rightarrow \quad 3x^2 - 2x - 2 = -1$

$\Rightarrow \quad 3x^2 - 2x - 1 = 0$

$\Rightarrow \quad (3x + 1)(x - 1) = 0$

$\Rightarrow \quad x = -\frac{1}{3}, 1$

2. Four persons can hit a target correctly with probabilities $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}$ and $\frac{1}{8}$ respectively. If all hit at the target independently, then the probability that the target would be hit, is:

A. $\frac{1}{192}$ B. $\frac{25}{32}$

C. $\frac{7}{32}$ D. $\frac{25}{192}$

Ans. B

Sol:

Given the probability of hitting a target independently by four persons are respectively

$P_1 = \frac{1}{2}, P_2 = \frac{1}{3}, P_3 = \frac{1}{4}$ and $P_4 = \frac{1}{8}$

Let four persons are A, B, C, D. Probability of Hitting target = 1 – (None of four person Hit the target)

$= 1 - P(\bar{A}).\,P(\bar{B}).P(\bar{C}).P(\bar{D})$

$= 1 - \frac{1}{2}\cdot\frac{2}{3}\cdot\frac{3}{4}\cdot\frac{7}{8}$

$= \frac{25}{32}$

3. The integral $\int \sec^{2/3} x\ \operatorname{cosec}^{4/3} x\ dx$ is equal to:

(Here C is a constant of integration)

A. $-3\cot^{-1/3} x + C$ B. $-\frac{3}{4}\tan^{-4/3} x + C$

C. $-3\tan^{-1/3} x + C$ D. $3\tan^{-1/3} x + C$

Ans. C.

Sol: Let

$$I = \int \sec^{2/3} x \operatorname{cosec}^{4/3} x\, dx$$

$$I = \int \frac{dx}{(\sin x)^{4/3}(\cos x)^{2/3}}$$

$$I = \int \frac{dx}{\left(\frac{\sin x}{\cos x}\right)^{4/3}.\cos^2 x}$$

$$I = \int \frac{\sec^2 x\, dx}{(\tan x)^{4/3}}$$

put $\tan x = t \Rightarrow \sec^2 x\, dx = dt$

$$I = \int \frac{dt}{t^{4/3}} = \frac{t^{-1/3}}{(-1/3)} + C$$

$$I = \frac{-3}{(\tan x)^{1/3}} + C$$

4. If the function $f : R - \{1, -1\} \to A$ defined by $f(x) = \frac{x^2}{1 - x^2}$, is surjective, then A is equal to:

A. R–[–1, 0) B. R–(–1, 0)

C. [0, ∞) D. R–{–1}

Ans. A

Sol: Given that

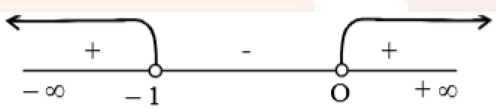
$$f(x) = \frac{x^2}{1 - x^2}$$

$$y = \frac{x^2}{1 - x^2}$$

$$\Rightarrow y - x^2 y = x^2$$

$$\Rightarrow x^2 = \frac{y}{1 + y}$$

$$\Rightarrow x = \sqrt{\frac{y}{1 + y}} \quad \Rightarrow \frac{y}{1 + y} \geq 0$$

$\Rightarrow$ 

Range of y is R – [–1, 0)

For surjective function codomain = Range

∴ A is R – [–1, 0)

5. For any two statements p and q, the negation of the expression $p \vee (\sim p \wedge q)$ is:

A. $p \leftrightarrow q$ B. $p \wedge q$

C. $\sim p \wedge \sim q$ D. $\sim p \vee \sim q$

Ans. C

Sol:

Given that,

$=\sim(p \vee (\sim p \wedge q))$

$= \sim p \wedge (p \vee \sim q)$

$= (\sim p \wedge p) \vee (\sim p \wedge q)$

$= c \vee (\sim p \wedge \sim q)$

So,

$= \sim p \wedge \sim q$

6. If the tangent to the curve, $y = x^3 + ax - b$ at the point (1, –5) is perpendicular to the line, $-x+y+4=0$, then which one of the following point lies on the curve?

A. (2, –1) B. (–2, 1)

C. (–2, 2) D. (2, –2)

Ans. D

Sol: Given that

$y = x^3 + ax - b$

(1, – 5) lies on curve

$\therefore -5 = 1 + a - b$

$\Longrightarrow a - b = -6 \qquad \ldots(1)$

$\frac{dy}{dx} = 3x^2 + a$

Slope of tangent at $(1, -5)$

$\Rightarrow \frac{dy}{dx} = 3 + a$

This tangent is perpendicular to $-x + y + 4 = 0$

$\therefore (3 + a)(1) = -1$

$\Longrightarrow a = -4 \qquad \ldots(2)$

By (1) & (2) a = – 4, b = 2

So, eq^n. of curve $y = x^3 - 4x - 2$

(2, –2) lies on this curve

7. Let $\vec{\alpha} = 3\hat{i} + \hat{j}$ and $\vec{\beta} = 2\hat{i} - \hat{j} + 3k$. If $\vec{\beta} = \vec{\beta}_1 - \vec{\beta}_2$, where $\vec{\beta}_1$ is parallel to $\vec{\alpha}$ and $\vec{\beta}_2$ is perpendicular to $\vec{\alpha}$, then $\vec{\beta}_1 \times \vec{\beta}_2$ is equal to:

A. $\frac{1}{2}\left(-3\hat{i} + 9\hat{j} + 5k\right)$ B. $\frac{1}{2}\left(3\hat{i} - 9\hat{j} + 5k\right)$

C. $-3\hat{i} + 9\hat{j} + 5k$ D. $3\hat{i} - 9\hat{j} - 5k$

Ans. A

Sol: Given that

$\vec{\alpha} = 3\hat{i} + \hat{j}$ and $\vec{\beta} = 2\hat{i} - \hat{j} + 3k$

$\vec{\beta}_1$ is parallel to $\vec{\alpha}$

$\therefore \vec{\beta}_1 = \lambda\vec{\alpha}$

$\Rightarrow \vec{\beta}_1 = \lambda\left(3\hat{i} + \hat{j}\right)$

Given that $\vec{\beta} = \vec{\beta}_1 - \vec{\beta}_2$

$\Rightarrow \vec{\beta}_2 = \vec{\beta}_1 - \vec{\beta}$

$\Rightarrow \vec{\beta}_2 = \lambda\left(3\hat{i} + \hat{j}\right) - \left(2\hat{i} - \hat{j} + 3\hat{k}\right)$

$\Rightarrow \vec{\beta}_2 = \hat{i}\left(3\lambda - 2\right) + \hat{j}\left(\lambda + 1\right) - 3\hat{k}$

Also given that $\vec{\beta}_2$ is perpendicular to $\vec{\alpha}$

$\therefore \vec{\beta}_2.\vec{\alpha}$

$\Rightarrow 3\left(3\lambda - 2\right) + \left(\lambda + 1\right) = 0$

$\Rightarrow \lambda = \frac{1}{2}$

So, $\vec{\beta}_1 = \frac{3}{2}\hat{i} + \frac{1}{2}\hat{j}$ and $\vec{\beta}_2 = \frac{-1}{2}\hat{i} + \frac{3}{2}\hat{j} - 3\hat{k}$

$$\vec{\beta}_1 \times \vec{\beta}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3/2 & 1/2 & 0 \\ 1/2 & 3/2 & -3 \end{vmatrix} = \frac{1}{2}\left(-3\hat{i} + 9\hat{j} + 5\hat{k}\right)$$

8. If one end of a focal chord of the parabola, $y^2 = 6x$ is at (1, 4), then the length of this focal chord is:

A. 24 B. 20

C. 25 D. 22

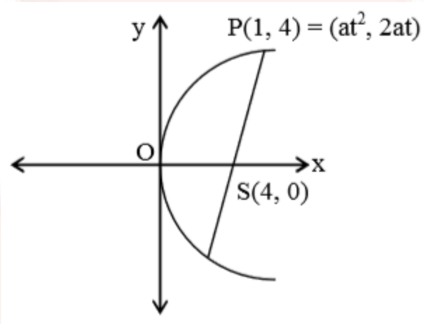
Ans. C

Sol:

Equation of given parabola is $y^2 = 16x$, its focus is (4, 0).

Parabola $y^2 = 16x$

$\{4a = 16 \Longrightarrow a = 4\}$



One end $(at^2, 2at) = (1, 4)$

$\Longrightarrow 2at = 4$

$\Longrightarrow 2(4)\,t = 4$

$\Longrightarrow t = 1/2$

Length of focal chord

$$= a\left(t + \frac{1}{t}\right)^2 = 4\left(\frac{1}{2} + 2\right)^2 = 25$$

9. If a tangent to the circle $x^2 + y^2 = 1$ intersects to coordinate axes at distinct points P and Q, then the locus of the mid-point of PQ

A. $x^2 + y^2 - 2xy = 0$

B. $x^2 + y^2 - 2x^2y^2 = 0$

C. $x^2 + y^2 - 4x^2y^2 = 0$

D. $x^2 + y^2 - 16x^2y^2 = 0$

Ans. C.

Sol:

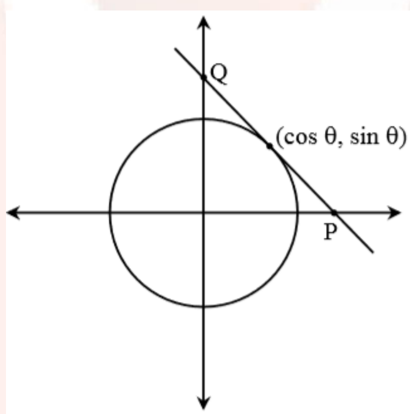
Equation of given circle is $x^2 + y^2 = 1$, then equation of tangent at the point $(\cos\theta, \sin\theta)$ on the given circle is

$x\cos\theta + y\sin\theta = 1$ …(i)

[∵ Equation of tangent at the point $P(\cos\theta, \sin\theta)$ the circle $x^2 + y^2 = r^2$ is $x\cos\theta + y\sin\theta = rI$]

Let the equation of tangent is $x\cos\theta + y\sin\theta = 1$ co-ordinates of P and Q are

$P\left(\frac{1}{\cos\theta}, 0\right)$ and $Q\left(0, \frac{1}{\sin\theta}\right)$



Let mid-point of P and Q is (h, k)

so, $h = \dfrac{\dfrac{1}{\cos\theta} + 0}{2}$ and $k = \dfrac{0 + \dfrac{1}{\sin\theta}}{2}$

$\Rightarrow \cos\theta = \dfrac{1}{2h}$ and $\sin\theta = \dfrac{1}{2k}$

squaring and adding we get

$\dfrac{1}{4h^2} + \dfrac{1}{4k^2} = 1$

$\therefore$ locus $\dfrac{1}{4x^2} + \dfrac{1}{4k^2} = 1$

$\Rightarrow x^2 + y^2 - 4x^2y^2 = 0$

10. A committee of 11 members is to be formed from 8 males and 5 females. If m is the number of ways the committee is formed with at least 6 males and n is the number of ways the committee is formed with at least 3 females, then:

A. n = m –8 B. m = n = 78

C. m = n = 68 D. m + n = 68

Ans. B

Sol:

Since there are 8 males and 5 females. Out of these 13 members committee of 11 members is to be formed.

m = no. of ways the committee is formed with at least 6 males.

$= {}^8C_6 \times {}^5C_5 + {}^8C_7 \times {}^5C_4 + {}^8C_8 \times {}^5C_3 = 78$

n = no. of ways the committee is formed with atleast 3 female

$= {}^8C_8 \times {}^5C_3 + {}^8C_7 \times {}^5C_4 + {}^8C_6 \times {}^5C_5$

= 10 + 40 + 28 = 78

$\Longrightarrow$ m = n = 78

11. A plane passing through the points (0, –1, 0) and (0, 0, 1) and making an angle $\frac{\pi}{4}$ with the plane *y*– z + 5 = 0, also passes through the point:

A. $\left(\sqrt{2},1,4\right)$ B. $\left(-\sqrt{2},-1,-4\right)$

C. $\left(-\sqrt{2},1,-4\right)$ D. $\left(\sqrt{2},-1,4\right)$

Ans. A

Sol:

Let the equation of plane is

ax + by + cz = d

Since plane (i) passes through the points (0, -1, 0) and (0, 0, 1), then

-b = d and c = d

∴ Equation of plane becomes ax – dy + dz = d

Given that angle b/w them is π/4

$$\cos\theta = \left|\frac{\vec{a}.\vec{b}}{|\vec{a}||\vec{b}|}\right|$$

$$\Rightarrow \frac{1}{\sqrt{2}} = \frac{|0-1-1|}{\sqrt{a^2+1+1}\sqrt{0+1+1}}$$

$$\Rightarrow a^2+2=4$$

$$\Rightarrow a=\pm\sqrt{2}$$

$\therefore$ eq^n. of plane $\pm\sqrt{2}\,x-y+z=1$

Now for – ve sign

$$-\sqrt{2}\left(\sqrt{2}\right)-1+4=1$$

$\therefore$ $\left(\sqrt{2},1,4\right)$ satisfy the eq^n. of plane.

12. If the line $y = mx + 7\sqrt{3}$ is normal to the hyperbola $\frac{x^2}{24} - \frac{y^2}{18} = 1$, then a value of m is:

A. $\frac{\sqrt{5}}{2}$ B. $\frac{\sqrt{15}}{2}$

C. $\frac{2}{\sqrt{5}}$ D. $\frac{3}{\sqrt{5}}$

Ans. C

Sol:

Given equation of hyperbola, is

$$\frac{x^2}{24} - \frac{y^2}{b^2} = 1$$

Since, the equation of the normals of slope m to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, are given by

Equation of normal of hyperbola in slope form is

$$y = mx \pm \frac{m\left(a^2 + b^2\right)}{\sqrt{a^2 - b^2 m^2}}$$

$$\therefore\ 7\sqrt{3} = \frac{42m}{\sqrt{24 - 18m^2}}$$

$$\Rightarrow\ 72 - 54m^2 = 36m^2$$

$$\Rightarrow\ 72 - 90m^2$$

$$\Rightarrow\ m^2 = \frac{72}{90} = \frac{4}{5}$$

$$\Rightarrow\ m = \pm\frac{2}{\sqrt{5}}$$

$$m = \frac{2}{\sqrt{5}}$$

13. If the standard deviation of the numbers –1, 0, 1, k is $\sqrt{5}$ where k > 0, then k is equal to:

A. $2\sqrt{\frac{10}{3}}$ B. $4\sqrt{\frac{5}{3}}$

C. $2\sqrt{6}$ D. $\sqrt{6}$

Ans. C

Sol:

Given observations are -1, 0, 1 and k.

Also, the standard deviation of these four observations = $\sqrt{5}$

$$\text{S.D.} = \sqrt{\frac{1}{n}\Sigma x_i^2 - (\bar{x})^2}$$

$$\text{Now mean } \bar{x} = \frac{-1+0+1+k}{4} = \frac{\cancel{-1}+0\cancel{+1}+k}{4}$$

$$\bar{x} = \frac{k}{4}$$

Given that S.D. = $\sqrt{5}$

$$\Rightarrow \sqrt{5} = \sqrt{\frac{1}{4}(1+0+1+k^2) - \frac{k^2}{16}}$$

$$\Rightarrow 5 = \frac{2+k^2}{4} - \frac{k^2}{16}$$

$$\Rightarrow 5 = \frac{4(2+k^2)-k^2}{16}$$

$$\Rightarrow 5 = \frac{8+4k^2-k^2}{16}$$

$$\Rightarrow 80 = 8 + 4k^2 - k^2$$

$$\Rightarrow 3k^2 = 72$$

$$\Rightarrow k^2 = 24$$

$$\Rightarrow k = \pm 2\sqrt{6}$$

$$k = 2\sqrt{6} \quad (\because k > 0)$$

14. Let $S=\{\theta\in[-2\pi,2\pi]:2\cos^2\theta+3\sin\theta=0\}$. Then the sum of the elements of S is:

A. $\frac{5\pi}{3}$ B. $\frac{13\pi}{6}$

C. π D. 2π

Ans. D

Sol: Given that

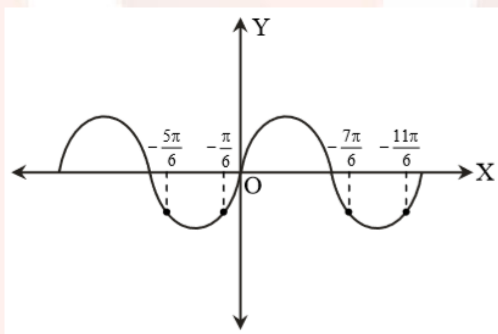
$2\cos^2\theta+3\sin\theta=0$

$\Rightarrow\ 2(1-\sin^2\theta)+3\sin\theta=0$

$\Rightarrow\ 2\sin^2\theta-3\sin\theta-2=0$

$\Rightarrow\ (2\sin\theta+1)(\sin\theta-2)=0$

$\Rightarrow\ \sin\theta=-\frac{1}{2}$



in $\theta\in[-2\pi,2\pi]$

$\Rightarrow\ \theta=-\frac{5\pi}{6},\frac{-\pi}{6},\frac{7\pi}{6},\frac{11\pi}{6}$

Sum of all roots $=\frac{-5\pi-\pi+7\pi+11\pi}{6}$

$=2\pi$

15. The solution of the differential equation $x\frac{dy}{dx}+2y=x^2\ (x\neq 0)$ with $y(1)=1$, is:

A. $y=\frac{3}{4}x^2+\frac{1}{4x^2}$ B. $y=\frac{4}{5}x^3+\frac{1}{5x^2}$

C. $y=\frac{x^3}{5}+\frac{1}{5x^2}$ D. $y=\frac{x^2}{4}+\frac{3}{4x^2}$

Ans. D

Sol: The given differential equation is

$x\frac{dy}{dx}+2y=x^2\left(x\neq 0\right)$

$\Rightarrow \frac{dy}{dx}+\frac{2y}{x}=x$

$I.F.=e^{\int \frac{2}{x}dx}=e^{\log_e x^2}=x^2$

$\therefore$ Solution is

$\Rightarrow yx^2=\int x^2.x\,dx$

$\Rightarrow yx^2=\frac{x^4}{4}+C$

at $y(1)=1\Rightarrow 1=\frac{1}{4}+C$

$\Rightarrow C=\frac{3}{4}$

$\Rightarrow yx^2=\frac{x^4}{4}+\frac{3}{4}$

$\Rightarrow y=\frac{x^2}{4}+\frac{3}{4x^2}$

16. The value of $\int_0^{\pi/2}\frac{\sin^3 x}{\sin x+\cos x}dx$ is:

A. $\frac{\pi-2}{4}$ B. $\frac{\pi-1}{4}$

C. $\frac{\pi-2}{8}$ D. $\frac{\pi-1}{2}$

Ans. B

Sol:

The property of definite integral

$\int_a^b f(x)dx=\int_a^b f(a+b-x)dx$

$$I = \int_0^{\pi/2} \frac{\sin^3 x}{\sin x + \cos x}\,dx \qquad \ldots(i)$$

$$I = \int_0^{\pi/2} \frac{\sin^3\left(\frac{\pi}{2} - x\right)}{\sin\left(\frac{\pi}{2} - x\right) + \cos\left(\frac{\pi}{2} - x\right)}\,dx$$

$$I = \int_0^{\pi/2} \frac{\cos^3 x}{\cos x + \sin x}\,dx \qquad \ldots(ii)$$

Adding (i) and (ii) we get

$$\Rightarrow\ 2I = \int_0^{\pi/2} \left(\frac{\sin^3 x + \cos^3 x}{\sin x + \cos x}\right) dx$$

$$\Rightarrow\ 2I = \int_0^{\pi/2} \frac{(\sin x + \cos x)(\sin^2 x + \cos^2 x - \sin x \cos x)}{(\sin x + \cos x)}\,dx$$

$$\Rightarrow\ 2I = \int_0^{\pi/2} (1 - \sin x \cos x)\,dx$$

$$\Rightarrow\ 2I = \int_0^{\pi/2} \left(1 - \frac{1}{2}\sin 2x\right) dx$$

$$\Rightarrow\ 2I = \int_0^{\pi/2} \left(x + \frac{\cos 2x}{4}\right)_0^{\frac{\pi}{2}}$$

$$\Rightarrow\ 2I = \left(\frac{\pi}{2} - \frac{1}{4}\right) - \left(\frac{1}{4}\right) = \frac{\pi}{2} - \frac{1}{2}$$

$$\Rightarrow\ I = \left(\frac{\pi - 1}{4}\right)$$

17. The value of

$\cos^2 10^\circ - \cos 10^\circ \cos 50^\circ + \cos^2 50^\circ$ is:

A. 3/2 B. $\frac{3}{2}(1 + \cos 20^\circ)$

C. $\frac{3}{4} + \cos 20^\circ$ D. 3/4

Ans. D

Sol: Given that

$\cos^2 10° - \cos 10° \cos 50° + \cos^2 50°$

$= \frac{1}{2}\left[2\cos^2 10° - 2\cos 10° \cos 50° + 2\cos^2 50°\right]$

$= \frac{1}{2}\left[\left(1+\cos 20°\right) - \left(\cos 60° + \cos 40°\right) + \left(1 + \cos 100°\right)\right]$

$= \frac{1}{2}\left[2 - \cos 60° + \cos 20° + \left(\cos 100° - \cos 40°\right)\right]$

$= \frac{1}{2}\left[2 - \frac{1}{2} + \cos 20° + 2\sin 70° \sin\left(-30°\right)\right]$

$= \frac{1}{2}\left[\frac{3}{2} + \cos 20° - \sin 70°\right]$

$= \frac{1}{2}\left[\frac{3}{2} + \cos 20° - \sin\left(90° - 20°\right)\right]$

$= \frac{3}{4}$

18. Let α and β be the roots of the equation $x^2 + x + 1 = 0$. Then for $y \neq 0$ in R, $\begin{vmatrix} y+1 & \alpha & \beta \\ \alpha & y+\beta & 1 \\ \beta & 1 & y+\alpha \end{vmatrix}$ is equal to:

A. $y\left(y^2 - 1\right)$ B. $y^3 - 1$

C. y^3 D. $y\left(y^2 - 3\right)$

Ans. C

Sol: The given equation is $x^2 + x + 1 = 0$

Root of eq^n. $x^2 + x + 1 = 0$ are α and β

$$\alpha, \beta = \frac{-1 \pm \sqrt{1-4}}{2} = \frac{-1 \pm i\sqrt{3}}{2}$$

$\Rightarrow \alpha = \omega, \beta = \omega^2$ (complex cube root of unity)

$$\Delta = \begin{vmatrix} y+1 & \omega & \omega^2 \\ \omega & y+\omega^2 & 1 \\ \omega^2 & 1 & y+\omega \end{vmatrix}$$

$R_1 \to R_1 + R_2 + R_3$

$$\Rightarrow \Delta = \begin{vmatrix} y & y & y \\ \omega & y+\omega^2 & 1 \\ \omega^2 & 1 & y+\omega \end{vmatrix} \left(\because 1 + \omega + \omega^2 = 0\right)$$

$$\Rightarrow \Delta = y\begin{vmatrix} y & y & 1 \\ \omega & y+\omega^2 & 1 \\ \omega^2 & 1 & y+\omega \end{vmatrix}$$

$\Delta = y\left(y^2\right)$

$\Delta = y^3$

19. Let $f(x) = 15 - |x - 10|; x \in R$. Then the set of all vales of *x*, at which the function, $g(x) = f(f(x))$ is not differentiable, is:

A. {10} B. {10, 15}

C. {5, 10, 15} D. {5, 10, 15, 20}

Ans. C

Sol:

Given function is f(x) = 15 - |x - 10|, x ∈ R and g(x) = f(fx))

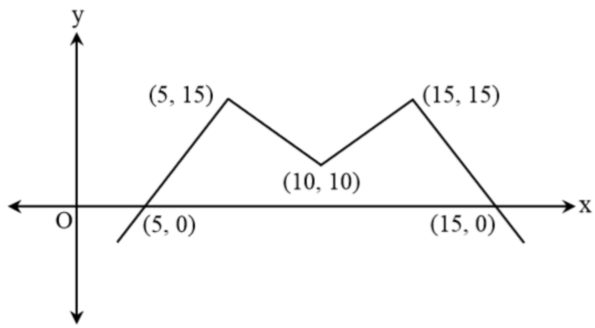
The given equation is

f(x) = 15 – |x – 10|

g(x) = f [f(x)] = 15 – | f(x) – 10 |

= 15 – | 15 – | x – 10 | – 10 |

= 15 – | 5 – | x – 10 ||



$\therefore$ g(x) is not differentiable at x = 5, 10, 15

20. All the point in the set $S = \left\{\frac{\alpha + i}{\alpha - i} : \alpha \in R\right\}\left(i = \sqrt{-1}\right)$ lie on a:

A. circle whose radius is $\sqrt{2}$.
B. circle whose radius is 1.
C. straight line whose slope is –1.
D. straight line whose slope is 1.

Ans. B

Sol: The given set $S = \left\{\frac{\alpha + i}{\alpha - i} : \alpha \in R\right\}\left(i = \sqrt{-1}\right)$

Let $\frac{\alpha + i}{\alpha - i} = z$

$\Rightarrow \left|\frac{\alpha + i}{\alpha - i}\right| = |z|$

$\Rightarrow |z| = 1$

$\Rightarrow$ Circle of radius $= 1$

21. The area (in sq. units) of the region $A = \{(x, y) : x^2 \le y \le x + 2\}$ is:

A. $\frac{9}{2}$ B. $\frac{13}{6}$

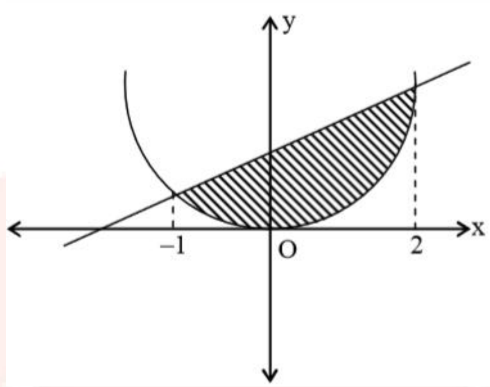
C. $\frac{10}{3}$ D. $\frac{31}{6}$

Ans. A

Sol:

Given region is $A = \{(x, y) : x^2 \le y \le x + 2\}$

Now, the region is shown in the following graph.



For intersecting points

$x^2 \le y \le x + 2$

$x^2 = y;\ y = x + 2$

$\Rightarrow\ x^2 = x + 2$

$\Rightarrow\ x = 2, -1$

So, area $= \int_{-1}^{2} \left\{(x + 2) - x^2\right\} dx = \dfrac{9}{2}$

22. Let the sum of the first n terms of a non-constant A.P., a_1, a_2, a_3, ……….. be $50n + \dfrac{n(n-7)}{2}A$, where A is a constant. If d is the common difference of this A.P., then the ordered pair (d, a_{50}) is equal to:

A. (50, 50 + 46A) B. (50, 50 + 45A)

C. (A, 50 + 45A) D. (A, 50 + 46A)

Ans. D

Sol:

The formula of sum of first n terms of AP, ie, $S_n = \dfrac{n}{2}[2a + (n-1)d]$

Given that the sum of the first n terms

$$S_n = 50n + \frac{n(n-7)}{2}A$$

$$T_n = S_n - S_{n-1}$$

$$T_n = 50n + \left(\frac{n(n-7)}{2}\right)A - 50(n-1) - \left(\frac{(n-1)(n-8)}{2}\right)A$$

$$= 50 + \frac{A}{2}\left[n^2 - 7n - n^2 + 9n - 8\right]$$

$= 50 + A(n – 4)$

Now, $d = T_n – T_{n–1}$

$= 50 + A(n – 4) – 50 – A(n – 5) = A$

and $T_{50} = 50 + 46A$

$(d, A_{50}) = (A, 50 + 46A)$

23. If the function *f* defined on $\left(\frac{\pi}{6}, \frac{\pi}{3}\right)$ by $f(x) = \begin{cases} \frac{\sqrt{2}\cos x - 1}{\cot x - 1}, & x \neq \frac{\pi}{4} \\ k, & x = \frac{\pi}{4} \end{cases}$

is continuous, then k is equal to:

A. $\frac{1}{\sqrt{2}}$ B. 2

C. 1 D. $\frac{1}{2}$

Ans. D

Sol:

Given function is $f(x) = \begin{cases} \frac{\sqrt{2}\cos x - 1}{\cot x - 1}, & x \neq \frac{\pi}{4} \\ k, & x = \frac{\pi}{4} \end{cases}$

$\because$ Function f(x) is continuous, so it is continuous at $x = \frac{\pi}{4}$.

$$\lim_{x\to\frac{\pi}{4}}\left(\frac{\sqrt{2}\cos x-1}{\cot x-1}\right)=f\left(\frac{\pi}{4}\right)$$

$$\Rightarrow \lim_{x\to\frac{\pi}{4}}\frac{\sqrt{2}\cos x-1}{\cot x-1}=k$$

using L-Hospital Rule

$$\Rightarrow \lim_{x\to\frac{\pi}{4}}\frac{\sqrt{2}\left(-\sin x\right)}{-\operatorname{cosec}^2 x}=k$$

$$\Rightarrow \sqrt{2}\left(\frac{1}{\sqrt{2}}\right)^3=k$$

$$\Rightarrow k=\frac{1}{2}$$

24. Let $\sum_{k=1}^{10} f\left(a+k\right)=16\left(2^{10}-1\right)$, where the function f satisfies $f(x + y) = f(x)\,f(y)$ for all natural numbers x, y and $f(1) = 2$. Then the natural number 'a' is:

A. 2 B. 4

C. 3 D. 16

Ans. C

Sol:

Given $f(1)=2$ and $f(x+y)=f(x)\cdot f(y)$

at $x=1,\ y=1 \Rightarrow f(2)=f(1)\cdot f(1)=2^2$

$x=2,\ y=1 \Rightarrow\ f(3)=f(2)\cdot f(1)=2^3$

……………………

……………………

$f(n)=2^n$

Now $\sum_{k=1}^{10} f(a+k) = 16(2^{10}-1)$

$\Rightarrow f(a+1)+f(a+2)+.....+f(a+10) = 16(2^{10}-1)$

$\Rightarrow 2^{a+1}+2^{a+2}+.....+2^{a+10} = 16(2^{10}-1)$

$\Rightarrow 2^{a}+\left[2^{1}+2^{2}+.....+2^{10}\right] = 16(2^{10}-1)$

$\Rightarrow 2^{a}\left[\frac{2(2^{10}-1)}{2-1}\right] = 16(2^{10}-1)$

$\Rightarrow 2^{a+1} = 16$

$\Rightarrow a = 3$

25. If $f(x)$ is a non-zero polynomial of degree four, having local extreme points at $x = -1, 0, 1$; then the set $S = \{x \in R : f(x) = f(0)\}$ contains exactly:

A. four rational numbers.
B. two irrational and two rational numbers.
C. two irrational and one rational number.
D. four irrational numbers.

Ans. C

Sol:

The non-zero four degree polynomial f(x) has extremum points at x = -1, 0, 1 so we can assume $f'(x) = a(x + 1)(x - 0)(x - 1) = ax(x^2 - 1)$ where, a is non-zero constant.

Four degree polynomial function f(x) have local extreme points at x = –1, 0, 1

$\therefore\ f'(x) = \lambda(x+1)(x-0)(x-1) = \lambda(x^3 - x)$

$\Rightarrow\ f(x) = \lambda\left(\frac{x^4}{4} - \frac{x^2}{2}\right) + K$

Now, $f(x) = f(0)$

$\Rightarrow\ \frac{x^4}{4} - \frac{x^2}{2} = 0$

$\Rightarrow\ x = 0, \pm\sqrt{2}$

Two irrational and one rational number.

26. Let p, q $\in$ R. If $2-\sqrt{3}$ is a root of the quadratic equation, x^2 + px + q = 0, then:

A. $q^2 + 4p + 14 = 0$
B. $p^2 - 4q + 12 = 0$
C. $q^2 - 4p - 16 = 0$
D. $p^2 - 4q - 12 = 0$

Ans. D

Sol:

If one root of equaion

$x^2 + px + q = 0$ is $2-\sqrt{3}$

then other root will be $2+\sqrt{3}$

$\therefore$ equation $x^2 - 4x + 1 = 0$

So, sum of roots = -p = 4 ⇒ p = -4

and product of roots = q = 4 – 3 ⇒ q = 1

Now, from options $p^2 - 4q - 12 = 16 - 4 - 12 = 0$

27. If

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \cdot \ldots\ldots\ldots\ldots \cdot \begin{bmatrix} 1 & n-1 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 78 \\ 0 & 1 \end{bmatrix},$$

then the inverse of $\begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix}$ is:

A. $\begin{bmatrix} 1 & -12 \\ 0 & 1 \end{bmatrix}$ B. $\begin{bmatrix} 1 & 0 \\ 12 & 1 \end{bmatrix}$

C. $\begin{bmatrix} 1 & -13 \\ 0 & 1 \end{bmatrix}$ D. $\begin{bmatrix} 1 & 0 \\ 13 & 1 \end{bmatrix}$

Ans. C

Sol:

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}\ldots\ldots\begin{bmatrix} 1 & n-1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 78 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 1+2+3+\ldots.+n-1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 78 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow 1+2+3+\ldots.+(n-1) = 78$$

We know that,

$$\Rightarrow \frac{n(n-2)}{2} = 78$$

$$\Rightarrow \frac{n^2-2n}{2} = 78$$

$$\Rightarrow n^2 - 2n = 78 \times 2$$

$$\Rightarrow n^2 - 2n = 156$$

$$\Rightarrow n = 13$$

Now, inverse of $\begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix}$ i.e. $\begin{bmatrix} 1 & 13 \\ 0 & 1 \end{bmatrix}$

is $\begin{bmatrix} 1 & -13 \\ 0 & 1 \end{bmatrix}$

28. Slope of a line passing through P(2, 3) and intersecting the line, $x + y = 7$ at a distance of 4 units from P, is:

A. $\frac{1-\sqrt{7}}{1+\sqrt{7}}$ B. $\frac{\sqrt{7}-1}{\sqrt{7}+1}$

C. $\frac{1-\sqrt{5}}{1+\sqrt{5}}$ D. $\frac{\sqrt{5}-1}{\sqrt{5}+1}$

Ans. A

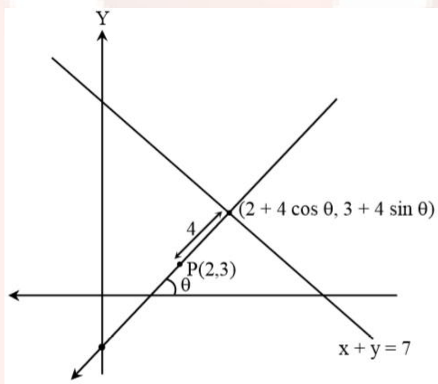
Sol:

The distance of a point (x_1, y_1) from the line ax + y + c = 0 is

$$d = \left|\frac{ax_1 + by_1 + c}{\sqrt{a^2 + b^2}}\right|.$$

Let any point on the line is $P(2 \pm 4\cos\theta, 3 \pm 4\sin\theta)$

it also lie on line x + y = 7



$\therefore\ (2 \pm 4\cos\theta) + (3 \pm 4\sin\theta) = 7$

$\Rightarrow\ (\sin\theta + \cos\theta) = \pm\frac{1}{2}$

$\Rightarrow\ (\sin\theta + \cos\theta)^2 = \frac{1}{4}$

$\Rightarrow\ 1 + \sin 2\theta = \frac{1}{4}$

$\Rightarrow\ \sin 2\theta = -\frac{3}{4}$

$\Rightarrow\ \frac{2\tan\theta}{1 + \tan^2\theta} = -\frac{3}{4}$

$\Rightarrow\ 3\tan^2\theta + 8\tan\theta + 3 = 0$

$\Rightarrow\ \tan\theta = \frac{-8 \pm 2\sqrt{7}}{6} = \frac{8 - 2\sqrt{7}}{6} = \frac{(1 - \sqrt{7})^2}{1 - 7} = \frac{1 - \sqrt{7}}{1 + \sqrt{7}}$

29. If the line, $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-2}{4}$ meets the plane, $x + 2y + 3z = 15$ at a point, P, then the distance of P from the origin is:

A. 9/2 B. $2\sqrt{5}$

C. $\sqrt{5}/2$ D. 7/2

Ans. A

Sol:

Equation of given plane is

x + 2y + 3z = 15

Line $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-2}{4} = k(\text{say})$ any point on this line P(2k + 1, 3k – 1, 4k + 2)

This point P lies on plane x + 2y + 3z = 15

∴ (2k + 1) + 2(3k – 1) + 3(4k + 2) = 15

⟹ 20k + 5 = 15

$\Longrightarrow 20k = 10$

$\Rightarrow$ 1/2 $\therefore$ $P\left(2, \frac{1}{2}, 4\right)$

Distance of P from origin is

$= \sqrt{4 + \frac{1}{4} + 16} = \frac{9}{2}$

30. If the fourth term in the Binomial expansion of $\left(\frac{2}{x} + x^{\log_8 x}\right)^6 (x > 0)$ is 20×8^7. Then a value of x is:

A. 8^3 B. 8

C. 8^{-2} D. 8^2

Ans. D

Sol:

Given binomial $\left(\frac{2}{x} + x^{\log_8 x}\right)^6 (x > 0)$

Since, general term in the expansion of $(x + a)^n$ is $T_{r+1} = {}^nC_r x^{n-r} a^r$

$\Rightarrow T_4 = 20 \times 8^7$

$\Rightarrow {}^6C_3 \left(\frac{2}{x}\right)^3 \left(x^{\log_g x}\right)^3 = 20 \times 8^7$

$\Rightarrow \frac{160}{x^3} x^{3\log_8 x} = 20 \times 8^7$

$\Rightarrow x^{3\log_8 x - 3} = 8^6$

$\Rightarrow x^{\log_2 x - 3} = 8^6 = 2^{18}$

$\Rightarrow \log_2\left(x^{\log_2 x - 3}\right) = \log_2 2^{18}$

$\Rightarrow (\log_2 x - 3)(\log_2 x) = 18$

Let $\log_2 x = t$

$\Rightarrow t^2 - 3t - 18 = 0$

$\Rightarrow t = 6, -3$

$\Rightarrow \log_2 x = 6 \quad \Rightarrow x = 2^6 = 8^2$

$\Rightarrow \log_2 x = -3 \quad \Rightarrow x = 2^{-3} = 1/8$